NOT MEASUREMENT SENSITIVE
MIL-I-7068A
15 December 1988
SUPERSEDING
MIL-Z-7068
15 December 1950

MILITARY SPECIFICATION

INGOT, ZINC ALLOY (FOR SHEET METAL FORMING DIES)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification addresses requirements for a zinc alloy used in the production of sand cast forming dies for sheet metal stamping operations.

1.2 Composition. Ingots shall be furnished in one of the following compositions (see 6.3):

Composition A (UNS Z35543)
Composition B (UNS Z35542)

1.2.1 Unless otherwise specified (see 6.2.1), ingot shall be composition A.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
MILITARY

MIL-N-3944	Nonferrous products (other than Aluminum, Magnesium, Copper, or their Alloys), Packaging and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 5314), Naval Air Engineering Center, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9650
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STANDARDS
MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-961	Military Specifications and Associated Documents, Preparation of

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are indicated as DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documenets cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	Commercial Packaging
ASTM E 536	Chemical Analysis of Zinc and Zinc Alloys

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

(Non-Government standards and other publicatios are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational sources.)

2.3 Order of precedence. In the event of a conflict between the contents of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical composition. Material shall conform to the chemical composition detailed in Table I.

TABLE I. Chemical composition.

Element	Composition A (percent)	Composition B (percent)
Copper	2.5 - 3.5	2.5 - 2.9
Aluminum	3.5 - 4.5	3.9 - 4.3
Magnesium	0.02 - 0.010	0.02 - 0.05
Iron	0.10 max	0.075 max
Lead	0.007 max	0.003 max
Cadmium	0.005 max	0.003 max
Tin	0.0005 max	0.001 max
Zinc	Remainder	Remainder

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3.2 Material quality. Ingots shall be in standard commercial shapes and sizes, and shall be free from dross, slag, and foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling.

4.2.1 Lot. A lot shall consist of all ingots poured from the same heat of the same composition submitted for inspection at one time.

4.2.2 Sampling for chemical analysis.

4.2.2.1 Ingot analysis. At least one ingot shall be tested from each lot and analyzed to determine conformance with 3.1. Two samples shall be obtained from the ingot by drilling through the ingot at one end and at the center. The drill bit shall be clean and shall not be lubricated during the drilling operation.Ingots not conforming to the requirements of 3.1 shall be rejected. Complete ingot analysis shall be maintained at the producers facility and shall be available to the procuring activity for review upon request.

4.2.2.2 Ingot identities indeterminable. When ingot identities of an inspection lot can not be determined, a specimen shall be taken from each ingot which is unidentified in accordance with 4.2.2.1.

4.2.4 Sampling for visual examination. Each ingot shall be examined to determine conformance to this specification with respect to markings.

4.3 Examination.

4.3.1 Visual examination. Each sample selected in accordance with 4.2.4 shall be examined visually to verify conformance to this specification.

4.3.2 Preservation, packing and marking for shipment. The preservation, packing and marking for shipment shall be examined to determine compliance with the requirements of Section 5.

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4.4 Test procedures.

4.4.1 Chemical analysis. Chemical analysis shall be made by ASTM E356, spectrochemical methods or other analytical method approved by the contracting activity.

5. PACKAGING

5.1 Preservation, packing and marking. Preservation and packing shall be in accordance with MIL-N-3944 or ASTM D3951, as specified (see 6.2). The levels of preservation and packing shall be as specified (see 6.2).

5.2 Marking. In addition to any special or other identification marking required by the contract (see 6.2), each ingot shall be marked in accordance with MIL-STD-129.

5.3 Packaging inspection. All packaging requirements shall be inspected in accordance with 4.3.2 to ensure compliance.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This alloy is intended for use in fabricating forming dies for sheet metal stamping under drop hammers and hydro presses. Composition B is a special purpose alloy of closer controlled chemistry.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (See 2.2).
c. Composition (see 1.2).
d. Quantity (see 3.4).
e. Levels of preservation, packing and marking (see 5.1).
f. Special markings (see 5.2).

6.3 UNS designation. The parenthetical unified numbering system (UNS) designations in section 1.2 are listed for informational purposes only and shall not be considered as substitutes unless they meet all the other requirements of this specification.

6.4 Subject term (keyword) listing.
Forming dies
Ingot
Zinc alloy

6.5 Title and identifier change. The title and identifer for this document have been changed to comply with the requirements of MIL-STD-961C, Paragraph 5.2.2, dated 20 May 1988.

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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